

Air-Quality Impact of Sustainable-Mobility Scenarios in Guimarães

OpenFOAM Hackathon Challenge — OFW21

CFD assessment of CO and NO_x at four sensitive receptors

Executive summary

Four mobility scenarios — the provided reference case, two fleet-electrification scenarios (S1: 20% of gas vehicles replaced by EVs; S2: 40%), and a Metro-Bus corridor scenario (S3) — were evaluated with a steady-RANS dispersion workflow on the real terrain of Guimarães. Pollutant concentrations of CO and NO_x were sampled at the four sensitive receptors defined by the challenge: the Public Hospital and the Martins Sarmiento, Francisco de Holanda and Santos Simões high schools. Relative to the reference, electrification lowers concentrations uniformly at every receptor (−20% for S1, −40% for S2), whereas the Metro Bus produces a strongly location-dependent benefit: about −41% at the Public Hospital — comparable to S2 — but only −5% at Santos Simões. The exact −20/−40% scaling of the electrification cases also serves as a consistency check on the modelling chain.

1. Objective

The challenge is to quantify how sustainable-mobility measures change urban air quality at four sensitive locations, using the provided reference case as the baseline and reporting the expected air-quality improvement of each scenario. The assessment metric is the pollutant concentration sampled at the receptor surfaces (ROI), reported per pollutant and compared against the reference both in absolute terms and as a percentage change.

2. Methodology

Domain and mesh. The provided terrain and region-of-interest building surfaces are recentred to the origin and meshed once with snappyHexMesh on a COST-732-style background domain; the same mesh, boundary conditions and solver settings are reused for every hour and every scenario, so only the emission input changes between scenarios and comparisons are fair.

Flow. The hourly wind field is computed with steady RANS (simpleFoam, $k-\epsilon$ with neutral atmospheric-boundary-layer wall functions). An all-round atmospheric inlet accepts the hour-varying wind direction; a first-order warm-up followed by a second-order restart keeps the solve stable on the steep terrain. The converged U , ϕ and turbulent viscosity are frozen and passed to the dispersion stage.

Dispersion. Each pollutant is transported as a separate passive scalar (scalarTransportFoam) on the frozen flow, with the road network carved into the ground as a dedicated 'streets' patch carrying a non-uniform fixed-gradient flux derived from the per-segment hourly emission factors. CO and NO_x are solved independently and reported separately. Concentrations are therefore the local traffic increment, with no regional background added.

Receptors and temporal strategy. Concentration is sampled as the area average over each of the four ROI receptor surfaces (surfaceFieldValue function objects). To keep the 24-hour problem tractable while preserving variability, ten representative hours were selected (maximin over wind speed/direction and emission totals) and run identically for all four scenarios; the statistics below are over those ten sampled hours. The whole sweep is orchestrated with a Snakemake workflow that keeps the large mesh decomposed end-to-end on the HPC system, reusing each hour's flow across all scenarios.

3. Scenarios

The three mobility scenarios are implemented purely as a scaling of the per-segment emission factors (the flow is identical across scenarios). S3 applies the reductions to the street segments listed in *road_ids_reduction.txt* — verified to map correctly onto the emission rows, and corresponding predominantly to the Circular Urbana ring together with one EN101 section, i.e. the bus corridor rather than the full EN101.

Scenario	Definition	Emission scaling
Reference	Provided traffic	x1.00 (baseline)
S1	20% of gas vehicles -> EV	x0.80, all segments
S2	40% of gas vehicles -> EV	x0.60, all segments
S3	Metro Bus (Guimaraes-Braga corridor)	x0.50 / x0.70 on the listed segments

4. Results

4.1 Baseline exposure

At baseline the most exposed receptor is Santos Simões (NO_x mean 6.5, peak 15.9 µg/m³), followed by Martins Sarmiento and the Public Hospital; Francisco de Holanda is the least exposed. NO_x is the pollutant of interest — CO is more than two orders of magnitude below its 8-hour guideline. Exposure peaks in the evening rush and at midday.

Receptor	CO mean	CO peak	NO _x mean	NO _x peak
Public Hospital	3.85	11.09	5.38	14.09
Francisco de Holanda HS	2.15	5.16	2.97	6.56
Martins Sarmento HS	3.76	9.70	5.52	12.60
Santos Simoes HS	4.46	11.23	6.47	15.91

Table 2. Reference concentrations (µg/m³), mean and peak over the ten sampled hours.

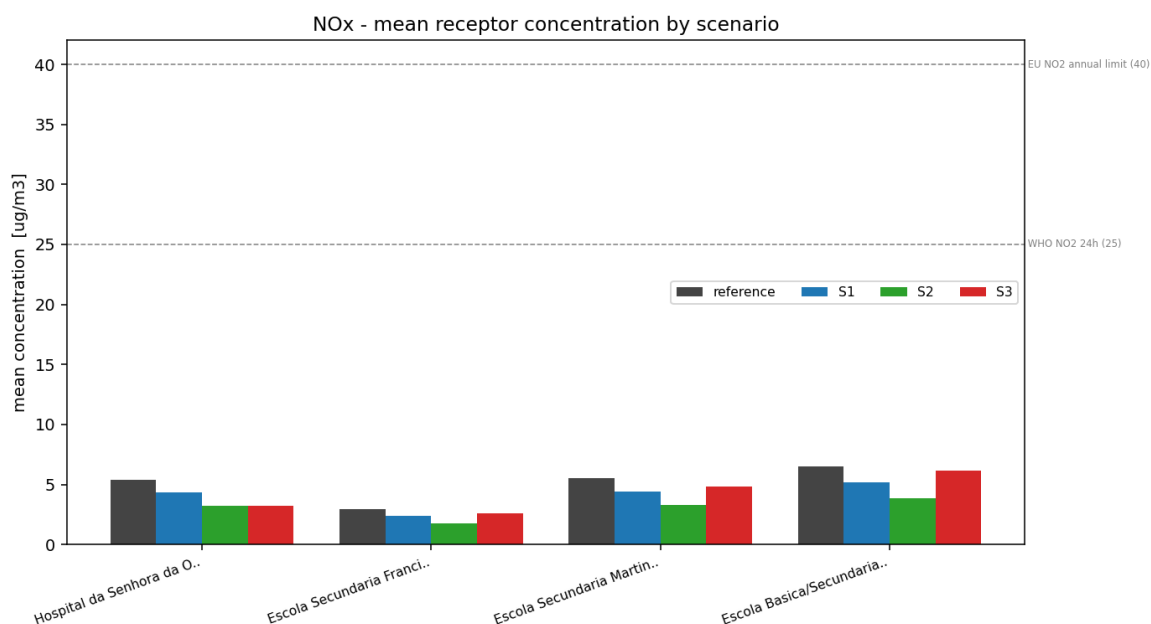


Figure 1. Mean NO_x at each receptor by scenario (dashed lines: EU/WHO NO₂ references; modelled NO_x is not ambient NO₂).

4.2 Scenario comparison

Electrification scales uniformly — every receptor falls by exactly –20% (S1) and –40% (S2), for both pollutants and for both the mean and the peak. The Metro Bus is spatially selective: it removes only about 19% of the city-wide emission total yet cuts the Public Hospital by roughly –41%, while barely affecting the off-corridor receptors.

CO — mean concentration (µg/m³) and change vs reference

Receptor	Reference	S1	S2	S3
Public Hospital	3.85	3.08 (-20.0%)	2.31 (-40.0%)	2.26 (-41.2%)
Francisco de Holanda HS	2.15	1.72 (-20.0%)	1.29 (-40.0%)	1.89 (-12.2%)
Martins Sarmento HS	3.76	3.01 (-20.0%)	2.25 (-40.0%)	3.25 (-13.5%)
Santos Simoes HS	4.46	3.57 (-20.0%)	2.68 (-40.0%)	4.24 (-4.9%)

NO_x — mean concentration (µg/m³) and change vs reference

Receptor	Reference	S1	S2	S3
Public Hospital	5.38	4.30 (-20.0%)	3.23 (-40.0%)	3.21 (-40.4%)
Francisco de Holanda HS	2.97	2.38 (-20.0%)	1.78 (-40.0%)	2.62 (-11.9%)

Martins Sarmento HS	5.52	4.42 (-20.0%)	3.31 (-40.0%)	4.84 (-12.3%)
Santos Simoes HS	6.47	5.18 (-20.0%)	3.88 (-40.0%)	6.14 (-5.1%)

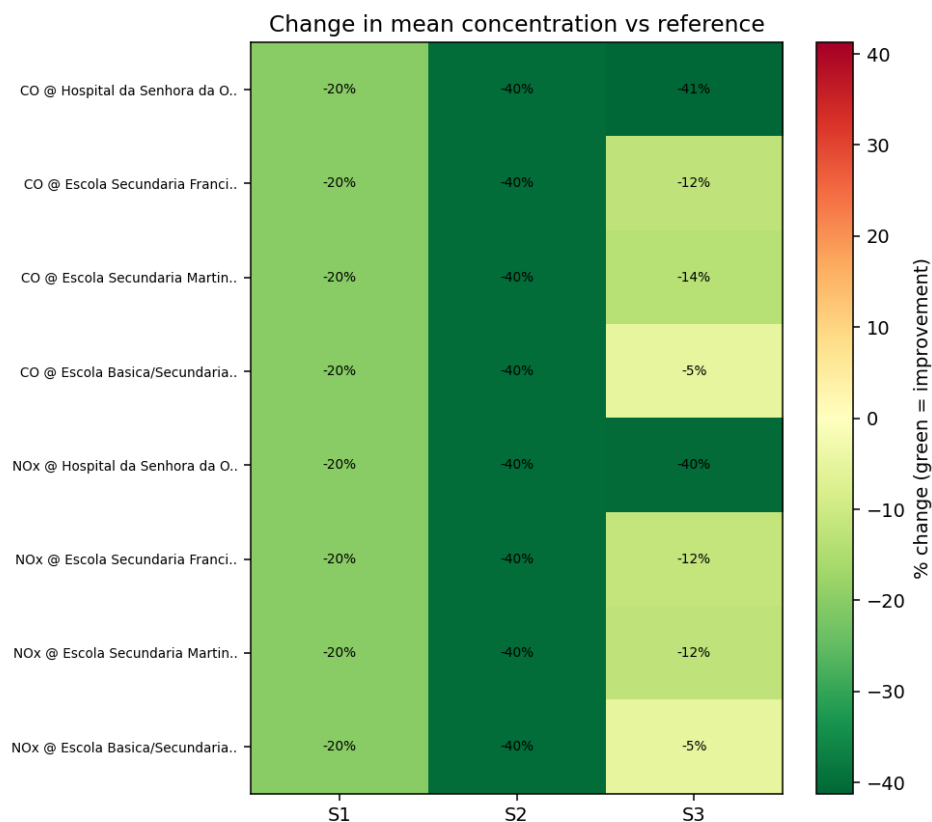


Figure 2. Change in mean concentration vs reference (green = improvement). S1/S2 uniform; S3 concentrated at the Hospital.

4.3 Spatial distribution

Placing the receptors on the Guimarães road network explains the scenario pattern. The S3 reduced corridor (blue) traces the Circular Urbana ring, which runs immediately next to the Public Hospital and far from Santos Simões — so the Metro Bus removes the Hospital's dominant local source while leaving the most-exposed school essentially untouched. Road width is proportional to NOx emission, marking the busiest source streets.

Guimaraes road network and NOx exposure at the four receptors (reference case)

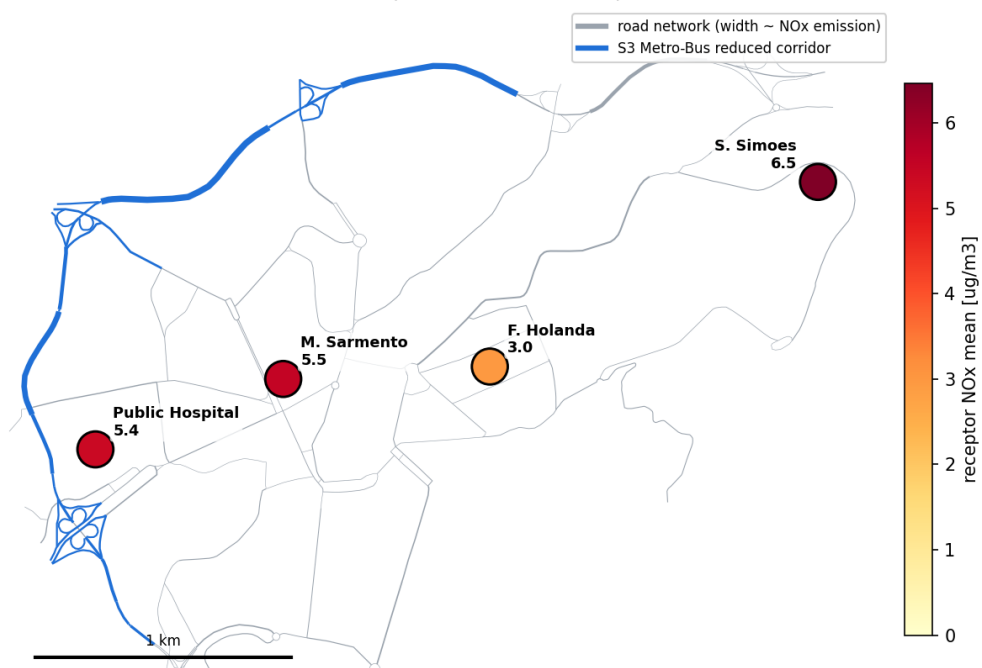


Figure 3. Road network (width ~ NOx emission), S3 reduced corridor (blue), and reference NOx at each receptor.

NOx at the four receptors by scenario (same colour scale)

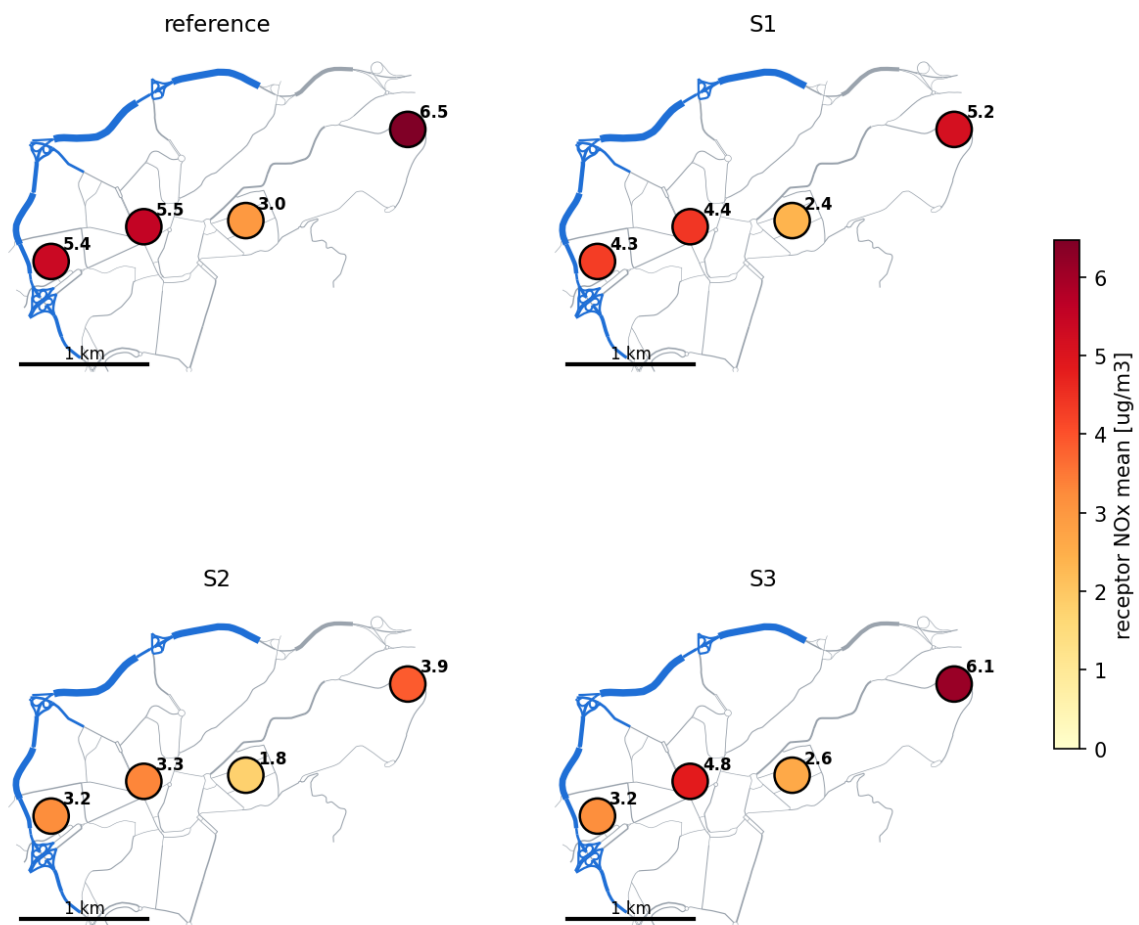


Figure 4. Receptor NOx by scenario on a shared colour scale: electrification lowers all sites uniformly; the Metro Bus (S3) acts mainly on the Hospital.

5. Discussion and conclusions

The three measures serve different objectives. S2 (40% electrification) is the strongest broad-based measure, lowering every sensitive receptor uniformly by 40%; S1 gives the same effect at half the magnitude. The Metro Bus is the most cost-effective intervention for the Public Hospital specifically — it achieves near-S2 benefit there while changing far less traffic overall, because the reduced corridor is the Hospital's dominant local source — but it does little for receptors away from the corridor and almost nothing for Santos Simões, which is the most exposed site at baseline.

The practical conclusion is that no single measure is best everywhere. A combined strategy — the Metro Bus to protect the Hospital and the central corridor, plus partial fleet electrification to bring down the off-corridor schools, above all Santos Simões — would deliver the most uniform city-wide improvement. If a single receptor must be prioritised, the choice of measure depends on which one: the bus for the Hospital, electrification for Santos Simões.

The exact –20% / –40% response of the electrification scenarios at every receptor is itself a useful result: dispersion is linear in emission for a frozen flow, so this uniform scaling confirms the emission mapping and the reuse of one flow field per hour are working as intended.

6. Limitations and future work

Statistics are over ten representative hours (the same hours for every scenario), so the 'mean' and 'peak' are sampled-hour figures rather than full 24-hour values; concentrations exclude regional background and modelled NO_x is not ambient NO₂, so the absolute levels should be read as relative traffic increments rather than compliance values. Natural extensions are: the full 24-hour cycle and a daily-aggregated exposure; the influence of the computational domain (a larger city4CFD domain with a porous vegetation canopy is already prepared); a combined Metro-Bus-plus-EV scenario; and spatial concentration maps of the corridor to complement the receptor numbers.